

Interventions per 1,000 Robot-Hours: Toward Standardized Reliability Evidence for Deployed Robot Fleets

Manuel Bulos
manuelbulos@gmail.com

July 2026

Abstract

Learning-based robots are entering commercial deployment, yet the robotics industry lacks a standardized way to measure, record, or report their field reliability. Intervention logs—records of humans pausing, resetting, teleoperating, or physically assisting a robot—are proprietary, unstructured, and mutually incompatible across vendors. As a consequence, no one can answer the question that operators, insurers, and certification bodies are beginning to ask: *how many human interventions does a robot fleet require per 1,000 robot-hours, and why?* This position paper argues that the robotics industry needs the equivalent of aviation’s flight data recorder and the autonomous-vehicle industry’s disengagement report, before regulation imposes one. We propose (i) a minimal set of fleet reliability metrics centered on the intervention rate per 1,000 robot-hours; (ii) an open recording schema that captures robot telemetry, policy actions, and human interventions on a shared clock; and (iii) an intervention taxonomy with root-cause labels. We release a reference implementation, including a simulated deployment that demonstrates cross-version regression detection from intervention telemetry alone. We close with a discussion of the incentive, privacy, and standardization challenges, and argue that shared failure evidence is a public good the field should build deliberately rather than reactively.

1 Introduction

Commercial deployments of learning-based robots crossed a threshold in 2025–2026: humanoid and manipulation systems now run production shifts in automotive plants and logistics facilities [Figure AI, 2026, Agility Robotics, 2025]. The public retrospectives of these pilots are notable both for what they report and for what they omit. Figure’s eleven-month deployment at BMW Spartanburg reports more than 1,250 hours of runtime and a task success rate near 98.7%—which, at production cadence, implies a failure roughly every 77 cycles, and on the order of hundreds of manual interventions over the program [Figure AI, 2026, iFactory, 2026]. The deployment’s stated goal of zero interventions per shift is not reported as achieved. Independent reviews of factory-floor humanoid programs find that intervention logs are treated as proprietary, that no standardized logging format exists, and that basic reliability questions—mean time between failures under production conditions, interventions per unit of operating time, root-cause distributions—cannot be answered from public information for any commercial robot fleet [KARP Research, 2026].

This is not a temporary reporting gap; it is a structural one. Robot vendors have every short-term incentive to keep failure data private, no shared schema in which to record it even if they wished to publish, and no agreed metric that would make two fleets comparable. The result is an

industry about to scale physical systems into human environments with less standardized reliability evidence than aviation had in the 1960s or the autonomous-vehicle (AV) industry had in 2015.

Two adjacent industries suggest what comes next. In aviation, flight data recorders became mandatory equipment decades ago, and standardized incident evidence is credited as a foundation of the field’s safety culture [International Civil Aviation Organization, 2016]. In the AV industry, California’s requirement that permit holders report *disengagements*—events where a safety driver takes control—created, whatever its flaws, the first public, cross-vendor, longitudinal reliability dataset in autonomy [California Department of Motor Vehicles, 2024]. Robotics has no equivalent of either instrument, while its regulatory context is moving quickly: ISO 25785-1, the first international safety standard specifically addressing industrial mobile robots with actively controlled stability (a category that includes humanoids), is in draft with publication expected around 2027 [International Organization for Standardization, 2025], and insurers underwriting robot deployments already demand documented risk assessments and audit trails as conditions of coverage [Humanoid Liability, 2026].

This paper takes a position: **the robotics industry should adopt a standardized, vendor-neutral format for recording and reporting field reliability evidence—now, while the installed base is small, rather than after regulation or a high-profile incident forces a worse version of the same requirement.** We make three concrete proposals and release a reference implementation.

2 The evidence gap

2.1 What deployments report today

Public reliability reporting from commercial robot deployments consists largely of vendor-selected aggregates: cumulative units shipped, cumulative tasks completed, and task success rates over unspecified conditions [KARP Research, 2026]. Success rates without exposure denominators (operating hours, cycles attempted) and without intervention accounting are not reliability evidence; a 98% success rate is compatible with both a robust system and one requiring constant supervision, depending on cycle time and recovery cost. The metrics that industrial buyers actually use to evaluate equipment—mean time between failures (MTBF), interventions per unit time, service burden, uptime—are either not recorded in comparable form or not disclosed [New Market Pitch, 2026].

2.2 Why the gap persists

Three forces sustain the gap. *Incentives*: in a competitive funding environment, no vendor benefits from unilaterally publishing failure rates. *Infrastructure*: even willing vendors have no shared schema; each team’s logging is an internal engineering artifact, typically retrofitted onto streaming pipelines that were designed for delivery rather than faithful recording [Robotics Tomorrow, 2026]. *Demand immaturity*: until recently, no external party had standing to demand better. This is changing: insurers now audit deployments and void coverage on mismatches between the assessed and actual application [Humanoid Liability, 2026], and conformity assessment against ISO 25785-1 will require operational records that today’s logging practices cannot produce [International Organization for Standardization, 2025].

2.3 The cost of the gap

The absence of standardized failure evidence is not only a transparency problem. It slows the field’s own progress. Datasets collected by skilled operators who rarely fail produce policies that never learn recovery behavior; deliberately captured failure and recovery episodes measurably improve policy robustness [EVS Intelligence, 2026]. Real-deployment failures—with the full telemetry context that preceded them—are among the scarcest and most valuable training signals in embodied AI, and today they are discarded or siloed at the moment of capture.

3 Proposal 1: A minimal metric set

We propose that deployed robot fleets be characterized by the following metrics, all computable from the schema in Section 4.

Metric	Definition	Unit
Intervention rate	Human interventions per 1,000 robot-hours	I/kh
MTBF	Robot-hours between interventions	hours
Autonomous success rate	Episodes completed without intervention	fraction
Recovered rate	Episodes completed only via human recovery	fraction
Mean recovery time	Mean duration of intervention events	seconds
Operator burden	Total operator time per 1,000 robot-hours	hours/kh

Table 1: Proposed fleet reliability metrics. Robot-hours are wall-clock hours in which the robot was tasked, including time under intervention.

The central quantity is the **intervention rate per 1,000 robot-hours** (I/kh). We argue for it, rather than task success rate, as the headline reliability metric for three reasons. First, it has an exposure denominator, making fleets of different sizes and duty cycles comparable. Second, it is behaviorally meaningful to operators: it translates directly into supervision staffing. Third, it has precedent: it is the robotics analogue of the AV industry’s miles-per-disengagement, a metric that, despite legitimate criticisms of gameability [California Department of Motor Vehicles, 2024], successfully created public, longitudinal, cross-vendor comparability where none existed.

An episode that requires human recovery must be classified as *recovered*, not *successful*. Conflating the two is the single most common distortion in current vendor reporting.

4 Proposal 2: A shared-clock recording schema

Reliability metrics are only as trustworthy as the records beneath them. We propose a minimal schema of four append-only streams, all timestamped against a single monotonic session clock:

1. **Frames:** proprioceptive state at a fixed rate—joint positions and velocities, end-effector pose, gripper force and aperture, and references to externally stored camera frames.
2. **Actions:** policy outputs, each tagged with the model version that produced it, the task phase, and optionally the policy’s own confidence.
3. **Interventions:** human take-over events with start and end times, intervention type, operator identity, task phase, and a root-cause label (Section 5).

4. **Episodes:** task attempts with outcome (*success, recovered, failure, aborted*) and the model version under which they ran.

Two design decisions matter more than the specific fields. First, the *shared clock*: interventions must be temporally alignable with the telemetry that preceded them, or root-cause analysis degenerates into anecdote. Retrofitted logging pipelines routinely fail this requirement [Robotics Tomorrow, 2026]. Second, *model-version tagging on every action*: learning-based systems change behavior with every policy update, and reliability claims that do not condition on model version are not reproducible. Version tagging is what turns a reliability log into a regression-detection instrument (Section 6).

The schema is deliberately storage-format agnostic; our reference implementation uses JSONL for auditability, and the streams map directly onto existing robot-learning dataset conventions such as RLDS and LeRobot [Ramos et al., 2021, Cadene et al., 2024], so that reliability recording and training-data capture can be the same act.

5 Proposal 3: An intervention taxonomy

Cross-fleet comparability requires that “a human stepped in” mean the same thing everywhere. We propose a two-level labeling: a closed set of intervention *types*, and an open, evolving set of root-cause labels.

Type	Definition
pause	Operator halted the robot; no state was changed.
reset	Task cycle was restarted from a defined initial state.
teleop	Operator took manual control of the robot to complete or correct the task.
physical	Operator physically altered the scene or the robot (cleared a jam, repositioned an object).
estop	Emergency stop, triggered by a human or a safety system.

Table 2: Proposed closed set of intervention types.

Root-cause labels (e.g., `grasp_slip`, `perception_miss`, `joint_fault`, `collision`) will inevitably be operator-dependent and incomplete. For this reason the schema records the telemetry window preceding every intervention, so that unsupervised clustering over failure signatures can recover root-cause structure even where labels are missing or inconsistent—an approach we demonstrate in the reference implementation.

6 Reference implementation

We release `robot-blackbox`, an open-source reference implementation of the schema and metrics.¹ It comprises a recording SDK whose API wraps an arbitrary control loop; the metric computations of Table 1; failure clustering over pre-intervention telemetry windows; cross-version regression detection; and an audit-oriented report generator.

To exercise the pipeline end-to-end without hardware, the implementation includes a simulated 6-DoF manipulation deployment with injectable failure modes whose telemetry signatures (gripper force collapse on slips, velocity variance on faults) are modeled on failure classes reported from real

¹<https://github.com/manuelbulos/robot-blackbox>

pilots. In a demonstration scenario, a fleet runs 200 cycles on a baseline policy version and 200 cycles on an updated version with a deliberately regressed grasping module. From intervention telemetry alone, the pipeline reports the update’s intervention rate as $1.8\times$ the baseline’s (8,766 vs. 4,782 per 1,000 robot-hours in the simulated fleet) and localizes the regression to the grasp phase via clustering. We emphasize that these figures are simulated and illustrate the *instrument*, not any real fleet; producing the first real cross-vendor baselines is precisely the point of standardizing the format.

7 Discussion

Incentives. No vendor will publish reliability evidence unilaterally, and this proposal does not ask them to. The near-term adopters are the parties who already need the records privately: operators who must staff supervision, insurers who must price deployments [Humanoid Liability, 2026], and vendors who must diagnose their own regressions. Public benchmarks can emerge later from anonymized aggregation, as they did in the AV industry only under regulatory compulsion [California Department of Motor Vehicles, 2024]. The window to shape the format voluntarily is before ISO 25785-1 conformity assessment and insurance requirements harden around ad-hoc alternatives [International Organization for Standardization, 2025].

Privacy and competition. Intervention records encode competitive information. The schema therefore separates the *format* (open) from the *data* (owned by the deploying parties), and aggregation for benchmarking requires explicit anonymization of task, site, and embodiment identifiers. The aviation precedent is instructive: standardized recorders coexist with confidential investigation regimes.

Gaming. Any headline metric invites optimization against it; AV disengagement counts were criticized on exactly this ground. Two properties of the proposed schema mitigate this: interventions are recorded with surrounding telemetry (making deletion detectable as coverage gaps on the shared clock), and the episode outcome taxonomy makes “success via human” unrepresentable as autonomous success.

Failure data as a training asset. Because the schema is compatible with robot-learning dataset formats [Ramos et al., 2021, Cadene et al., 2024], every recorded intervention is simultaneously a labeled failure-and-recovery episode—the training signal that demonstration datasets systematically lack [EVS Intelligence, 2026]. Reliability evidence and recovery-learning data are the same bytes; the industry currently throws both away.

8 Conclusion

Robotics is repeating a pattern aviation and autonomous driving have already lived through: capability arrives before evidence infrastructure, and evidence infrastructure arrives only when demanded from outside. The field has a short window in which the instrument can be designed by the people who understand robots, rather than imposed by those who do not. A shared clock, four streams, five intervention types, and one honest metric—interventions per 1,000 robot-hours—would be enough to start.

References

Agility Robotics. Agility robotics and GXO announce multi-year agreement. <https://www.agilityrobotics.com>, 2025. First formal RaaS deployment of humanoid robots in logistics.

Remi Cadene, Simon Alibert, Alexander Soare, Quentin Gallouedec, Adil Zouitine, and Thomas Wolf. LeRobot: State-of-the-art machine learning for real-world robotics in PyTorch. <https://github.com/huggingface/lerobot>, 2024.

California Department of Motor Vehicles. Autonomous vehicle disengagement reports. <https://www.dmv.ca.gov/portal/vehicle-industry-services/autonomous-vehicles/disengagement-reports/>, 2024.

EVS Intelligence. Embodied AI data collection: Teleoperation, sim-to-real, and data quality. <https://www.evsint.com/embodied-ai-data-collection-teleoperation-sim-to-real-2026/>, 2026.

Figure AI. F.02 contributed to the production of 30,000 cars at BMW. <https://www.figure.ai/news/production-at-bmw>, 2026. Deployment retrospective, BMW Group Plant Spartanburg.

Humanoid Liability. ISO 25785-1: Humanoid robot safety standards and liability implications. <https://humanoidliability.com/resources/iso-25785-humanoid-safety-standards/>, 2026.

iFactory. Humanoid robots at BMW Spartanburg: Figure 02 pilot lessons. <https://ifactoryapp.com/industries/automotive-manufacturing/humanoid-robots-bmw-spartanburg-figure-02-pilot-lessons>, 2026.

International Civil Aviation Organization. Annex 6 to the convention on international civil aviation: Operation of aircraft, 2016. Flight recorder carriage requirements.

International Organization for Standardization. ISO/AWI 25785-1: Robotics — safety requirements for industrial mobile robots with actively controlled stability — Part 1, 2025. Under development; working draft published May 2025.

KARP Research. Humanoid factory reality check. <https://souldriver.com.au/report/humanoid-factory-reality-check>, 2026. Independent review of factory-floor humanoid deployments, May 2026.

New Market Pitch. Humanoid robots in factories: For when? <https://newmarketpitch.com/blogs/news/humanoid-robotics-factories-when>, 2026.

Sabela Ramos, Sertan Girgin, Léonard Hussenot, Damien Vincent, Hanna Yakubovich, Daniel Toyama, Anita Gergely, Piotr Stanczyk, Raphael Marinier, Jeremiah Harmsen, Olivier Pietquin, and Nikola Momchev. RLDS: an ecosystem to generate, share and use datasets in reinforcement learning. arXiv:2111.02767, 2021.

Robotics Tomorrow. Why every robot teleoperation team eventually outgrows WebRTC. <https://www.roboticstomorrow.com/article/2026/05/why-every-robot-teleoperation-team-eventually-outgrows-webrtc/26597>, 2026.